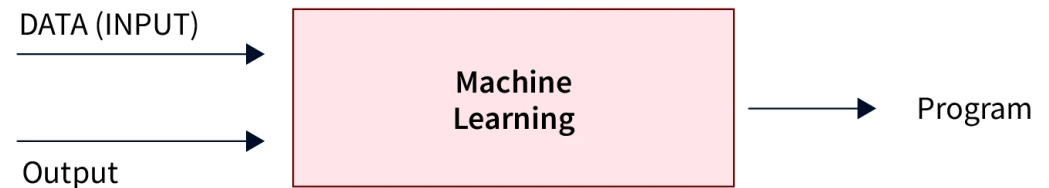


# INTRODUCTION TO MACHINE LEARNING

Moving from "Follow Instructions"  
to "Learn Patterns"

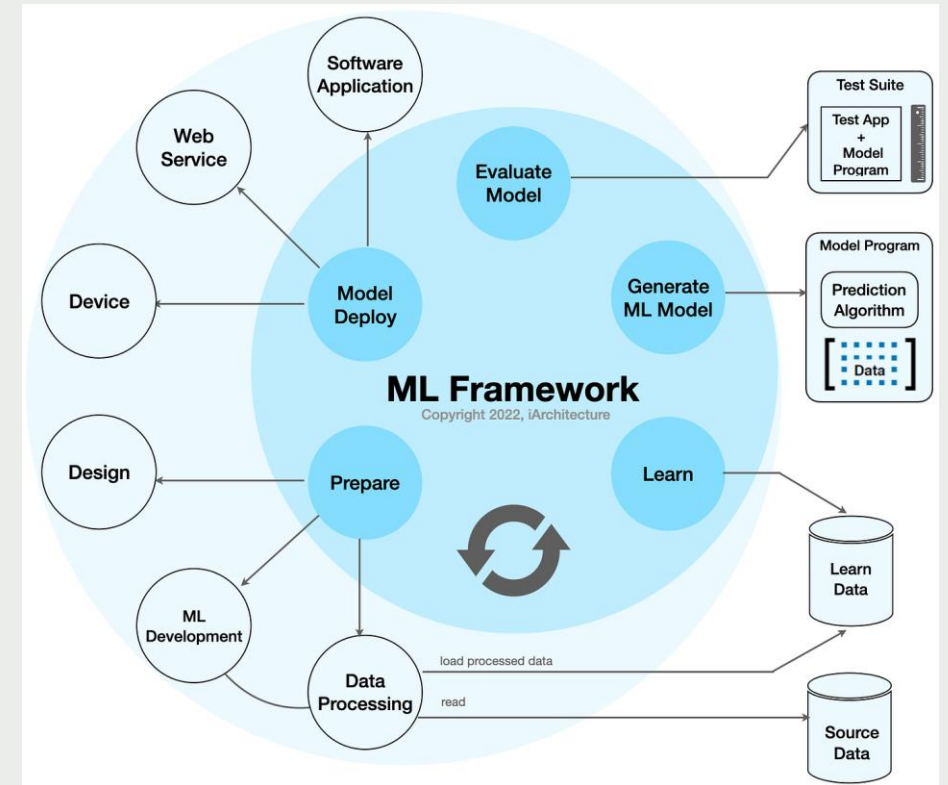
**Instructor:** Zaryab Ahmad Khan



SCALER  
*Topics*

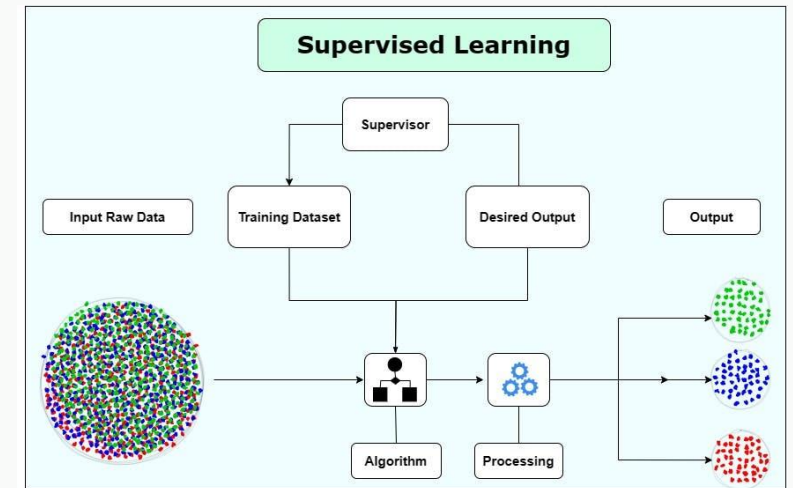
# The ML Landscape

- **Traditional Programming:** You write the rules (If/Else). The computer follows them.
- **Machine Learning:** You show the computer examples (Data + Answers). The computer writes the rules.
- **Analogy:** Teaching a child to recognize a cat by describing it (Traditional) vs. showing them 1,000 photos of cats (ML).



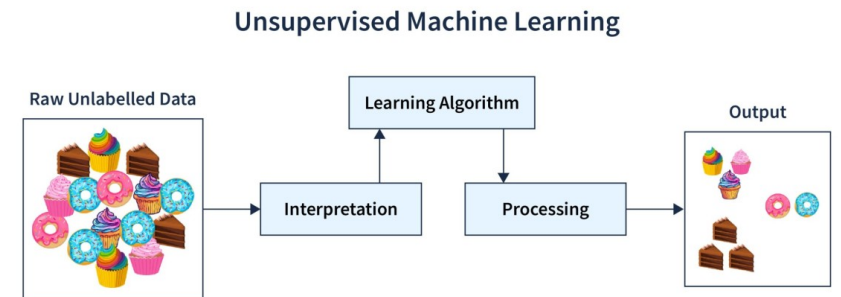
## Supervised Learning (The Guided Path)

- **Concept:** Learning with a "Teacher".
- **Data Type:** Labeled (Input + Target Answer).
- **Real-World Use:**
  - Predicting house prices based on square footage.
  - Identifying if a medical scan shows a tumor or not.



## Unsupervised Learning (The Explorer)

- **Concept:** Finding "Hidden Structure" without help.
- **Data Type:** Unlabeled (Input only).
- **Real-World Use:**
  - Customer Segmentation: Grouping shoppers by spending habits.
  - Anomaly Detection: Finding unusual bank transactions.



# Why Scikit-Learn?

- **Standardization:** Use the same "Fit/Predict" workflow for almost any model.
- **Efficiency:** Built on NumPy and SciPy for high performance.
- **Community:** The most documented ML library in the world.

